Part:	**1.4 – COMPOSITE MATERIALS**
Chapter:	**1412**
Title:	**DESCRIPTION OF DESIGN DATA FOR SANDWICH CORES**
Revision:	**1.0**
Date:	**15/12/2021**

1 INTRODUCTION

This chapter lists and describes the physical and mechanical properties required for stress analyses with core materials used in sandwich panels.

The statistical methods applicable calculate their design values from tests results data are presented in chapters 1480 “DERIVATION OF DESIGN DATA IN COMPOSITES.

Details on tests methods, on statistical methodologies, conventions and policies for presentation of data, are out of the scope of this document. See list of references for further information about these topics.

2 DEFINITIONS OF MATERIALS DESIGN DATA

2.1 IDENTIFICATION AND DESCRIPTION

Material identification and description shall be given in a similar manner as for other material types.

Short (informal) Name	
Provider and Commercial Name	
Material Designation(s) and Specification(s)	
Description	
Version	

2.2 PHYSICAL PROPERTIES

Symbol	***Units***	***Description***
c	mm	Cell size (*)
ρ	kg/m³	Density
MOL	ºC	Maximum Operational Limit, or maximum continuous service temperature without material irreversible degradation

2.3 CORE STIFFNESS

Due to the low in-plane stiffness of the honeycomb cores (compared with fibrous composite faces) and the “free” expansion in out plane direction, thermal expansion coefficients are not normally necessary for stress analysis.

<u>Elastic moduli</u>

Symbol	***Units***	***Description***
G_{13}	MPa	Shear modulus in longitudinal/though-thickness material plane (*)
G_{23}		Shear modulus in transverse/though-thickness material plane (*)
E_3		Elastic modulus in through-thickness direction
v	-	Poisson ratio (**)

These values shall be provided for the relevant ambient conditions as mean values and as B-basis values (natural dispersion of elastic moduli in cores is typically much higher than in FRP’s).

Test methods used for obtaining these properties shall be typically ASTM C273 (shear) and ASTM C365 (flatwise tension).

2.4 CORE STRENGTH

Symbol	***Units***	***Description***
F_{13}	MPa	Shear strength in longitudinal/though-thickness material plane (*)
F_{23}		Shear strength in transverse/though-thickness material plane (*)
F_{3t}		Tension strength in through-thickness direction
F_{3c}		Compression strength in through-thickness direction

These values shall be provided for the relevant ambient conditions at least as B-basis values.

Test methods used for obtaining these properties shall be typically ASTM C273 (shear) and ASTM C365 (flatwise tension and compression strengths)

(*) As-is for honeycomb cores. Foam cores shall be treated as isotropic and shear planes 13 and 23 will be treated indistinctly (one *G* shear modulus and one *Fsu* shear strength)

(**) Poisson ratio v only required for foam core (input for wrinkling failure analysis, see chapter §3510)

<u>*Remark*</u>

For the whole sandwich detail (including faces and core), using flat panels fully representative of the real part (materials, faces laminates, process, …), in order to perform the strength analysis with appropriate allowables, it is strongly recommended to test edgewise compression (ASTM C364) and/or 4-point-bending (AITM1-0018). Empirical parameters of formulations governing the sandwich failure modes related to local stability (e.g. wrinkling) shall be adjusted for the part with those data.

3 APPENDICES

OTHER ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
1	-	Ply direction parallel to fibres in unidirectional tape materials, and in warp direction in fabrics, as subscript
2	-	Ply direction normal to fibres in unidirectional tape materials, and in weft direction in fabrics, as subscript
3	-	Ply through-thickness direction, as subscript
EKDF / K_{EKDF}	-	Environmental Knock Down Factor to account for degradation of a mechanical property with material saturated in an ambient (temperature and humidity) different to the reference conditions of 20ºC (Room Temperature or RT) and dry material
K_B	-	B-basis Knock Down Factor for the minoring to the mean value of a mechanical property in order to get a safe design allowable value accounting for natural dispersion
TBC	-	To Be Confirmed

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[SAE] CMH-17	2012	COMPOSITE MATERIALS HANDBOOK
Ref. 2	[Airbus] PP0300181	Is. 2	Composite statistical method to determine B-basis value from small data sets"
Ref. 3	[Airbus] RP0412191	Is. 1	Normalization rules for composite properties

SOFTWARE

Program	***Version***	***Description***

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
1.0 15/12/2021	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods (TEPAP-TL4)	G. Baños 15/12/2021	Florian Glock 15/12/2021

Table 3. *Approval signatures table for last revision of this chapter*